Online Resource 1

The Review–Execution Gap in Agentic Software: Approval Scope and Recovery

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This supplement retains the archived formal, implementation, controlled-workload and historical integration details, and adds the prospective policy design and collection rules. The archived evidence layers, completed original prospective ledger and completed supplementary deployment-A evaluation remain separately identified. Sections, tables and figures prefixed S belong to this document. The complete original prospective ledger and result tables retain terminal failures and unknown coverage on the original planned denominator. Section S12 reports the separately frozen supplementary paired evaluation of deployment A; Section S13 reports an explicitly exploratory post hoc sensitivity view. Neither changes the original analysis.

Reading guide. For the execution contract and formal assumptions, start with Sections S2–S3. Sections S1 and S4–S7 preserve the controlled and historical evidence. Sections S8–S9 define the online design and reproducibility procedures; Section S11 contains the complete original results. Section S12 provides the complete supplementary A outcomes, provenance and reproduction steps; Section S13 records the sensitivity composition and results. The main manuscript presents four figures covering the concept, binding design space, paired online design and recovery/cost comparison. Figure S1 maps the evidence layers; Figure S3 retains the original recovery paths.

S1 Artifact, protocol, and reproduction

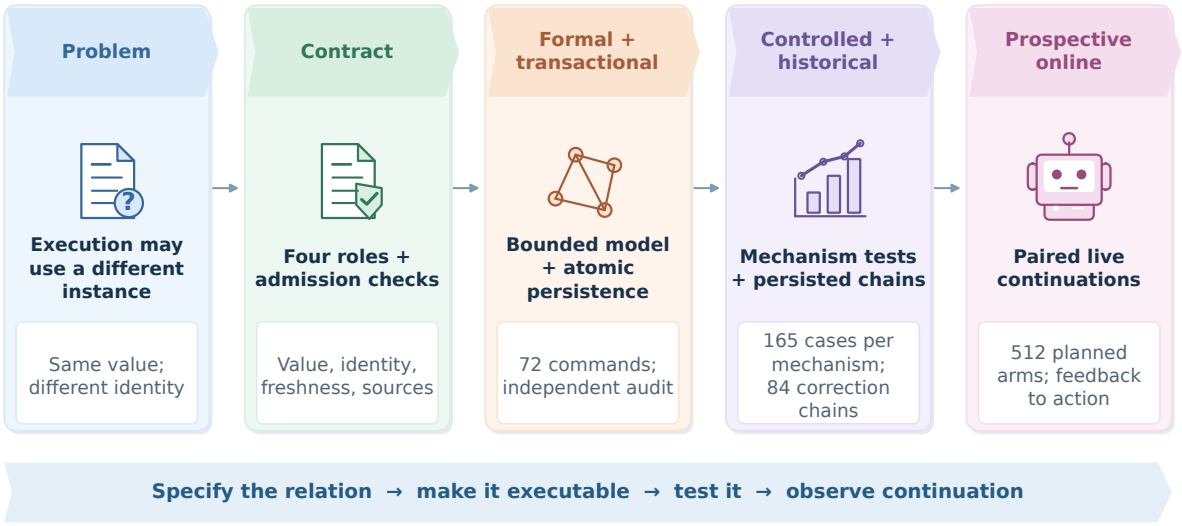


Figure S1: Evidence chain for trustworthy corrected execution. Distinct layers define continuity, make it executable, validate persistence and integration, then observe live agent continuation. Counts retain separate units: bounded commands, cases per mechanism, archived correction chains and planned paired trajectories.

S1.1 Three evidence layers

The reference implementation is fixed at commit `c6d512843c905cab6d8521dd8c914f7fb26d85ae` of `https://github.com/lhh666-6/auto-dete`, using `latest/code/implementation-fixed`. The tag `r31-jss-2026-09-13` retains formal models, observation witnesses, repaired code, and historical analyses. The historical v8 evidence remains under `r21-jss-2026-09-07-v8`. These layers are not substituted for one another when reporting counts or timings.

The new E1–E3 package is fixed at `https://github.com/lhh666-6/auto-dete/tree/2645e5e18c900ea91c9c980e44195dc71e410432/DKE-supplement`. Its `PROTOCOL.md` was drafted before outcome collection on 23 September 2026 and records pilot-driven amendments. It is a local prospective execution plan, not an externally registered protocol. Pilots remain in separately named directories and are excluded from final statistics. No hosted-model call was made in E1–E3.

Table S1: New experiment evidence and interpretation units. Paths are relative to the pinned `DKE-supplement` directory.

Layer	Primary records	Unit and limit
E1	`results/e1/inputs.json`; `results/e1/receipts.jsonl`	495 constructed executions; 165 per arm; seven distinct proposal values.
E2	`results/e2/browser-observations.jsonl`; screenshots and databases	60 scripted cases plus six replay setup confirmations; not a human study.
E3	`results/e3/*/raw.jsonl`; `summary.json`; `paired.json`	22,400 timed observations, excluding warmups; one machine.
Checks	`rejection-audit.json`; `verification.json`; `regression-receipt.json`	Expected rejection causes, counts, state digests, and test execution receipt.
Integrity	`artifact-manifest.json`; `source-integrity.json`	Content identities, not third-party attestation.

S1.2 Inputs, cases, and negative controls

E1 selects the first lexicographically ordered archived run containing a proposal in each of the G1/G2/D1 by B1–B4 cells; success labels are not a selection criterion. The twelve cells contain four distinct proposed values: 8, 100, the string `B-008`, and 9. Three additional controls exercise $2 \to 2.0$, $1 \to$ true, and nested integer/float changes. These values enter newly constructed form histories with fresh certificates and review records. This is not an end-to-end replay of the original tasks. Input records retain source paths, event identifiers, and hashes.

The eleven cases per input are Accept, Correction, multi-field admission, equal-valued candidate substitution, different-evidence substitution, stale authorization, replay, an unauthorized submitted value, and interruptions after CAS, before transition insertion, and before commit. The rich-context journal retains field, version, evidence, producer and proposal context, but its review projection excludes certificate identity, candidate identifier, and instance creation time. Its exact variant adds the reviewed-candidate link. Each successful three-field record is checked with six queries per field: proposed value, authorized value, reviewer, reviewed candidate, current value, and source version. The 18 answers per accepted case are query observations, not independent experiments.

E2 uses the JSON values 100, 2.5, and true. Each appears in ten interactions per path: Accept, Correction, legitimate reselection, candidate-request substitution, authorized-value substitution, principal substitution, stale state, replay, precommit interruption, and display-only substitution. The driver records DOM candidate identity, proposal, and authorized value before confirmation. Replay cases each require an initial confirmation, explaining the six additional setup calls. The float value 2.5 avoids JavaScript serialization normalizing an integral float to an integer. The

gate records a server-held review session using fixed simulated principal labels and a database separate from the fact database.

The rejection audit checks all 372 rejected attempts across E1 and E2, including their exact expected error categories or injected failpoints. No rejected attempt changes the measured fact-database digest. The baseline is taken after authorization preparation: already recorded review intent is not required to disappear when admission fails. This distinction also prevents the session database from being silently included in a fact-database atomicity claim.

S1.3 Execution environment and commands

The recorded environment is Intel Core i7-10875H, Windows 10 build 19045, CPython 3.11.16, SQLite 3.53.1, and SQLAlchemy 2.0.51. Browser orchestration uses Playwright and local Chrome. E3 uses a seeded random ordering of the two arms (seed 20260923), five warmups, and 200 timed observations per arm and cell. Admission uses SQLite DELETE/FULL; the storage-population measurement uses WAL/NORMAL followed by checkpointing. Background operating-system activity was not controlled.

The deposited README specifies dependency installation, reference-source resolution, and the separate browser driver environment. The following commands illustrate the entry points; the output directory must be fresh, and the reference checkout must be configured as described there. Performance should run after other tests have finished.

```
python -B experiment1.py --output results/replicate-e1
python -B experiment2.py --output results/replicate-e2
python -B experiment3.py --output results/replicate-e3
```

Pilot flags reduce coverage and cannot reproduce final denominators. The reference regression invocation covers its value-equality integration tests and conformance directory (85 passes); the supplementary test module records ten passes. The regression receipt transcribes the execution tool output rather than presenting itself as a pytest-generated JUnit report. The manifest records 61 pre-run measurement-source hashes; the reporting check found them unchanged. The original artifact manifest covers 3,068 files.

S1.4 Lossless storage packaging

The 100,000-transition full database is 141,062,144 bytes and is published as `results/e3/storage/storage-100000/full.db.gz`. The publication package includes `restore_large_database.py` and `publication-packaging.json`. Restoration checks the uncompressed SHA-256 against the original manifest; its verified value is

`a6ab45b86b3f198e8452b7adae756012807efe09ceb92d32c759cd9d2ffb816e`.

Compression changes transport size, not the database footprint used in the storage result. The restoration helper refuses to overwrite a differing existing file. Source and artifact manifests support reproducibility checks; they do not establish external validity or independent replication.

S2 Formal and implementation evidence

S2.1 Complete observation functions and witness construction

Main Section 3.3 summarizes the endpoint-indistinguishability witness. Online Resource 2, A9.1–A9.2, defines the observation domain, opaque-handle condition and local outcome, and states the T1a–T1c characterization. The following table and relational representation specify the executable projection used by the archived constructions.

Table S2: Observation functions used by the conditional characterization. "Derives" lists information available without consulting another class.

Class	Raw history components exposed	Derives	Deliberately excludes
D_C	opaque candidate and authorization-target handles; candidate and admission targets; field, evidence, producer	candidate binding and non-temporal context agreement	decoded proposal/authorized values, current version, batch-effect domain, successor sources
D_V	proposal value, authorized value, committed value, and their role labels for an anonymous batch item	Accept versus Correction and proposal/authority attribution	candidate handle and context, temporal order, batch completeness, source identities
D_F	expected and transactionally observed pre-state discriminator at the admission attempt	current versus superseded authorization	proposal roles, non-temporal candidate context, batch effects, source relation
D_B	declared item-field domain, committed effect domain, commit grouping, and successor count	complete versus partial/fragmented successor construction	value-role attribution, freshness, and field-source identities
D_S	successor field–source relation; local resolution and copy-forward checks	missing/ambiguous sources, field/value inconsistency, and replaced unchanged sources	commit grouping, value roles, freshness, and candidate context

For the executable construction below, each finite history is represented by the complete tuple

$$\widehat{h} = \langle r, C, A, B, v_{\text{exp}}, v_{\text{obs}}, F_{\text{decl}}, K, V_v, V_{v'}, S_v, S_{v'}, T \rangle, \tag{1}$$

containing the target, candidate and authorization registries, batch references, expected and observed pre-versions, declared fields, a commit sequence K, endpoint value and source maps, and a registry T of source-transition identifiers, fields, and values. Each commit step records a distinct successor identity and complete before/after value maps. The domain predicate checks unique candidate and authorization keys, resolvable batch references, schema-valid fields, total value maps, and a connected commit sequence from V_v to $V_{v'}$. Its committed effect domain is the union of the steps' actual value-change domains; its successor count is $|K|$. Neither is an independently assigned witness attribute.

Source observations expose each field's anchor count, reverse-resolution count in T, agreement of the resolved field/value with the successor, and exact prior-source retention for an unchanged field. Missing, duplicate, or inconsistent sources remain semantic failures rather than being excluded by the domain predicate. These are local source checks: the complete authorization–candidate–evidence chain belongs to the operational P6 checks defined below. Source observations hide intermediate commit grouping; freshness observes the admission-entry pre-state comparison. These explicit observation boundaries make it possible to compare atomic and fragmented histories with identical endpoints.

Online Resource 2, A9.2, states the observation-class characterization T1a–T1c and the reduced-projection relation; Main Section 3.3 summarizes the separate endpoint witness. The original candidate/context witness changes record context with a fixed candidate handle. The separate identity-isolation pair fixes the two-candidate registry and the non-identity observations, then changes the admitted candidate. The executable checker derives outcomes and observations from the complete tuple rather than storing either as an independent attribute.

S2.2 Evidence identity and operational enforcement details

Evidence identity is not content equality alone. The realization uses

$$EID(e) = \langle H(\text{bytes}(e)), L(r, f, u, \kappa) \rangle, \tag{2}$$

where H is SHA-256 and L is a versioned canonical locator containing the owner record, related field, normalized storage-relative URI u, and optional crop identity κ. The Alloy model abstracts content and locator as primitive domains. Its `certIdentityFacts` assigns distinct `CertId` atoms to distinct certificate objects, including objects with equal modeled content. Relating this abstraction to the implementation requires collision resistance of the deployed digest and integrity of the canonicalization and evidence store; it is not a proof or claim that SHA-256 is mathematically injective.

Full operational contract. Table S3 defines the operational properties referred to by the formal and implementation evidence. Main Section 3 defines the roles and obligations, Section 4 organizes the binding choices, and Section 5 describes the executable realization.

Table S3: Operational contract for a corrected authoritative transition.

Property	Requirement	Operational interpretation
P0	Direct-write exclusion	Candidate-writing capability cannot change authoritative values, versions, sources, transitions or authorizations.
P1	Policy-selected binding	Context requires equality of submitted and reviewed contexts; bound additionally requires the same immutable instance. The formal Full model checks the bound realization through canonical certificate identity.
P2	Context integrity	Record, field, persisted evidence content and locator, and expected predecessor agree across the item and certificate.
P3	Authorization integrity	The principal is allowed at the in-transaction check, and the authorization fixes the certificate and explicit committed value.
P4	Freshness	The expected predecessor equals the transactionally observed current version; stale or competing admissions are rejected.
P5	Successor integrity	One atomic successor has exact changed-field effects, an item–transition bijection and complete unchanged-field framing.
P6	Trace soundness	Every field resolves through its exact source transition to consistent authorization, candidate, evidence, producer and version relations.

These are paper-level obligation labels. Online Resource 2, A9 defines policy-relative satisfaction and maps these labels to the historical Alloy assertion names; source numbering is not a one-to-one match.

Admitted effects and no-op semantics. After initial coverage, concrete admissions require a nonempty value change under canonical JSON equality. For submitted items U, the admitted batch is $B = \{i \in U \mid i.x \neq_{\text{json}} V_v(i.f)\}$, with $B \neq \varnothing$. All items in B share one record and expected predecessor and have unique changed fields. A mixed submission admits exactly B; unchanged items retain their exact preceding source and create no new transition or authorization binding. Initial coverage establishes a complete value and source domain. The Alloy model additionally permits same-value admission for correspondence with the historical singleton model. A renewed review or re-attribution without a value change uses a separate review-event path, outside the concrete admission contract.

The trusted base includes the admission service, exercised transaction semantics, canonicalization, hashing, principal policy, and trusted lookup of content-immutable candidate records

Table S4: Mapping from failure-distinguishing information classes to bounded and concrete evidence.

Class	Safe/unsafe witness	Alloy support	Concrete support
D_C	matching / mismatched non-temporal candidate context; separate equal-valued identity-isolation pair	field, evidence-identity, evidence-owner, record, authorization-certificate, certificate-preexistence, and evidence-preexistence ablations	record, field, evidence, and certificate substitution cases
D_V	retained Correction / wrong attribution of the same final value	legal Correction witnesses and authorization-value ablation	singleton, all-field, and mixed Correction lifecycles with immutable candidate checks
D_F	current / superseded authorization	certificate-version and freshness ablations; stale-rejection assertions	stale replay, contending confirmation, and compare-and-swap rejection
D_B	atomic / partial or fragmented successor	transition-bijection ablation and whole-batch framing assertions	exact changed-field validation, failpoints, rollback, and concurrency cases
D_S	complete / missing, replaced, or duplicate source	source-anchor attack witnesses and P6 trace assertions	source-corruption catalogue and reverse-trace validation

preserved to consumption. A candidate-creation capability must not overwrite a retained candidate or its review linkage. This is a storage prerequisite separate from P0; the direct-write exclusion list alone does not establish it. Authentication, review presentation and evidence storage must supply the identities and content on which those checks operate. Rejection stutters relative to authoritative state immediately before the admission attempt; it need not erase evidence, candidate preparation or review intent persisted earlier.

Authorization timing and revocation The prototype evaluates the in-memory principal-role allow-list inside the database transaction, before the version compare-and-swap and any authority write. A revocation that completes before this check causes zero-write rejection. The allow-list lock protects only the policy lookup/update; it is not held for the remainder of the database transaction. Consequently, a revocation that overlaps an admission after its successful policy check does not cancel that in-flight commit. The implementation therefore provides commit-entry revalidation, not linearizable revocation against already admitted operations, distributed identity-provider semantics, or session invalidation. Deployments requiring those properties must place a transactional policy epoch or equivalent revocation discriminator inside the admission compare-and-swap.

Human authorization is not a truth oracle. The contract establishes who authorized which persisted candidate and value in which context, and whether the successor was constructed atomically. It cannot establish that the authorized value is factually correct.

S2.3 Information-class and persisted-state mappings

Table S4 connects each information class to its failure witness and executable evidence. Table S5 records how persisted relations are represented in the concrete/formal projection.

S2.4 Legal, unsafe, and preservation commands

Six legal-witness commands cover multi-field Accept, Correction, mixed Accept/Correction, same-value admission, initial snapshot, and singleton admission. Eleven paired ablations remove the conjuncts mapped in Table S4. Principal membership remains fixed; the unused `ABL_7` marker is excluded because host identity trust is an external assumption.

Three source-attack commands start from a legal Full prefix and then remove a source anchor, replace it, or add a duplicate source-version transition; each requires an incomplete

Table S5: Principal elements of the executable abstraction.

Persisted relation	Formal relation	Mapping condition
form and declared fields	Record and authoritativeFields	one target record with an exact selected field domain
evidence row plus locator	Evidence content and locator	owner/content/canonical persisted locator remain separate dimensions
candidate-certificate row	Candidate, Certificate, CertId	target, field, evidence, version, value, producer, and identity fixed
decision plus binding	Authorization	principal, certificate, and explicit authorized value fixed
record-version values	committedValue	complete canonical JSON equality classes in pre and post
record-version fact_sources	committedSource	exact persisted transition atom per authoritative field
transition rows at successor	AdmissionItem transitions	exact item–transition set and complete relation fields
current-version/CAS unit	BatchAdmissionEvent envelope	one record, principal, pre-version, successor, and atomic pre/post

trace. Fourteen assertions run per scope. Nine guard definitions and immediate consequences: P0/P1/P3/P4/P5, whole-batch effects, rejection framing, and two P6 checks. Five check invariant preservation, two singleton correspondences, rejection of an ablated attempt, and positive trace completeness under a well-formed pre-state.

The positive assertion requires a trace-complete predecessor and no stray transition to the successor version; otherwise the permissive base effect admits a post-state that fails `traceComplete`. It covers admitted fields in one step, not arbitrary histories or every carried-forward field. Both profiles include reachable initial and non-vacuous carry-forward witnesses for the precondition. Removing the post-state certificate binding changes UNSAT to SAT. These are separate checks of reachability, bounded preservation, and sensitivity, rather than fourteen independent proofs. Concrete enforcement additionally validates predecessor prefixes and uses transition uniqueness and in-transaction CAS revalidation.

S2.5 Scopes and interpretation

The S1 profile bounds principal structural domains at two records, three fields, six versions, up to six admission items/transitions for most commands, and four states/events. S2 increases representative bounds to three records, four fields, eight versions, up to eight admission items/transitions, and five states/events. Individual commands adjust Value, Evidence, or Transition bounds when a witness needs an extra atom. The artifact contains the executable scopes; this summary is not a replacement for them.

An UNSAT assertion means that the Analyzer found no counterexample within the encoded command scope. A SAT run means that it found the requested witness; the main text cites the Analyzer and solver references. Singleton contract/effect checks additionally guard the batch lift against silently changing the historical one-item semantics.

S2.6 Observation witnesses and finite scopes

The executable witness checker derives observations and outcomes from complete relational histories. Five class pairs test the necessity witnesses in Online Resource 2, A9.2 (T1a); a further pair isolates instance identity with a fixed two-candidate registry, and a legal control changes

one field while copying another field's exact source. The original source is `latest/code/formal-fixed/observation_witnesses.py`, with report `latest/evidence/r27-witness/observation-witness-report-v5.json`. The handoff also includes the repaired executable construction and 24 tests under `research/authorization-granularity-phase1-2026-10-06/formal/`.

Across the two scope profiles, the archived command manifest records 72 expected matches: 22 SAT and 14 UNSAT per profile. Sections S2.4–S2.5 define their commands, premises and scope interpretation. The independent projection outcomes and concrete checks below provide a separate check of the implementation-to-model boundary.

S2.7 Projection and database oracles

The projection adapter reads persisted pre/post state independently of the production planner and trace builder, maps JSON equality classes and identity relations into a fixed Alloy instance, and checks committed or rejected wrappers. Nine intended instances are SAT and twenty selected mapping mutants are UNSAT. A rejected wrapper additionally requires equal full-database digests. The separate 35-case catalogue has five legal cases and thirty rejection, corruption, schema-prevention, stateful-evolution, or oracle-mutant cases. Post-persistence corruption requires a changed digest and a trace diagnosis rather than zero writes.

Historical test snapshots have different counts: the initial reported suite has 354 passes, subsequent development has 377, and the r26 isolated export has 389, including optional-Git and revocation checks. These numbers describe their archived invocations. The new 85-test invocation is a selected regression run against the repaired reference source, not a claim to have rerun all 389 checks or to add 85 independent faults. Static-check and historical concurrency receipts remain in the original evidence layer.

S2.8 Canonical equality and witness-domain repairs

An earlier implementation inherited host-language equality that equated integer 2 with float 2.0 and integer 1 with Boolean true. A serialized value could change without obtaining a corresponding changed-field transition. The repair aligns the planner and reverse trace with canonical JSON equality already used by the independent oracle and formal projection. The resulting implementation consistently uses sorted-key, whitespace-minimal JSON serialization for change and value checks. Thirteen targeted regressions include unchanged controls, numeric/Boolean and nested changes, and no-op rejection; seven fail before the repair and all pass afterward. E1 adds explicit type controls using the repaired path.

The witness checker also separates the record schema domain from the declared item domain, allowing a legal change to one field while copying another field's source forward. This preserves the omission constructions without requiring every admission to change every field. These repairs and their receipts are versioned separately from v8; the cost grids in Section S4 measure the repaired source.

S3 Worked correction and implementation inspection

Consider a two-field record at version v: quantity is 90 with source t_q, and batch is `B-008` with source t_b. Candidate c_1 proposes quantity 100 for version v. A reviewer authorizes quantity 101 from c_1. The legal successor $v+1$ has quantity 101 from a new transition t_1, batch `B-008` from the exact previous source t_b, and an authorization naming both c_1 and 101. Candidate c_1 still contains 100. Canonical JSON equality distinguishes authorized integer `101` from floating serialization `101.0`, even if a domain treats their numeric magnitudes as equal. This is an explanatory construction, not an additional measured case.

A newly persisted equal-valued candidate c_2 can have the same value and retained review context while being a different instance. Under instance-level authorization, substituting c_2 for

c_1 must fail; under the declared context-level policy it may be admitted. A concurrent successor invalidates the expected predecessor. A proposed two-field admission cannot be split into two commits merely because the final endpoint is the same. Replacing the unchanged batch source with an equal-valued transition also loses the exact source relation.

In the online facade, successful commit consumes the authorization. Identical arguments and the same idempotency key retrieve the prior receipt without creating a transition; using a new key under that consumed grant is semantic replay and is rejected (Section S8). A domain-specific typed adapter would be an extension: normalization would need shared semantics across the planner, admission gate, trace verifier, independent oracle and formal projection, followed by renewed validation. Such an adapter is not an implemented or validated drop-in replacement for canonical JSON equality.

The following checklist locates the enforcement and observation points. It is an aid derived from the contract, not an independently validated development methodology.

Table S6: Design checklist derived from the contract and its five information classes. Adapt the probes to the target implementation and retain persisted evidence for each outcome.

Class	Retain and check	Failure probe and expected evidence
D_C	Exact candidate, target, evidence and producer; bind the decision at review and recheck context inside admission.	Substitute an equal-valued candidate from another context: reject without authority writes. Reload the binding to verify candidate identity.
D_V	Proposal, human-authorized value and committed value with separate roles; preserve the candidate and check the latter two values at admission.	Authorize 101 from proposal 100: commit 101, retain 100 and its identity. Reconstruction must expose false machine attribution.
D_F	Expected and observed predecessor discriminator; compare within the transaction and protect it against races.	Supersede the predecessor before replay: reject without changing the new current state. Competing admissions from one predecessor must not both commit.
D_B	Admitted field domain, actual effects and commit grouping; validate the planned successor and commit it atomically.	Change two fields; interrupt between writes. Recovery must expose all or none of that admission, never an intermediate or fragmented successor.
D_S	One resolving source per successor field; validate changed-field sources and exact unchanged-source retention before commit and on trace.	Keep values fixed but remove, duplicate or replace a source. Reject invalid construction; diagnose incomplete trace after persisted corruption.

The earlier SQLite demonstrator and its strengthened transactional variant agree with candidate-bound admission on seven of eight single-history cases. Equal-valued substitution is their only decision difference; a ninth paired-history check concerns distinguishability rather than a further admission category. The continuity rerun is deposited in `results/original-baseline-continuity`. It is separate from E1's richer-context three-arm design and is not added to the 495-execution denominator. In E1, extending the new journal with exact binding eliminates the observed decision and query-answer gap on the tested cases.

S4 Complete current-version cost grids

S4.1 Admission and materialization

The mechanism and materialization experiments each contain ten cells, two arms, and 200 observations per arm/cell: 4,000 timed observations each. Warmups and database-copy setup are excluded. The mechanism timer covers confirmation after adapter construction; the journal opens its sqlite3 connection in construction, whereas the reference ORM can connect lazily inside the call. Different schemas, certificate work, application features, and connection lifecycles confound attribution to an individual integrity check.

Table S7: Complete confirmation paths with pre-registered recognition candidates. Entries are p50 / p95 in milliseconds; 200 observations per arm/cell.

Fields	Changed	Reference	Exact journal
1	1	28.79 / 50.50	10.40 / 13.32
8	1	33.99 / 55.40	11.19 / 15.18
8	4	57.97 / 103.89	11.88 / 16.47
8	8	75.71 / 129.14	21.51 / 26.31
32	1	34.75 / 54.66	11.41 / 15.90
32	4	55.32 / 101.11	23.89 / 31.17
32	32	187.24 / 292.72	13.07 / 17.77
128	1	29.35 / 57.56	11.33 / 14.76
128	4	44.39 / 72.72	11.61 / 16.75
128	128	712.00 / 1039.83	22.09 / 30.53

The materialization experiment constructs a valid relational delta before timing, persists that delta in its reduced arm, and checks the final relational fingerprint after timing. The full arm uses manual-source fixtures, whereas the mechanism comparison uses pre-registered recognition candidates. Thus the two full-confirmation columns belong to different workloads and must not be subtracted across experiments. Random identifiers and timestamps also prevent a claim that paired database files are byte-identical; equivalence checks use the declared relational expectations.

Table S8: Full confirmation and prevalidated materialization. Entries are p50 / p95 in milliseconds; 200 observations per arm/cell.

Fields	Changed	Full confirmation	Materialization
1	1	23.23 / 33.76	9.97 / 14.63
8	1	23.87 / 65.83	10.40 / 15.05
8	4	36.68 / 57.24	11.48 / 19.46
8	8	50.23 / 97.50	11.38 / 17.22
32	1	27.00 / 45.75	11.00 / 16.52
32	4	44.23 / 97.98	17.31 / 23.66
32	32	122.68 / 200.00	13.89 / 22.61
128	1	29.10 / 48.08	11.08 / 17.21
128	4	38.33 / 87.89	11.34 / 17.19
128	128	427.59 / 621.20	23.50 / 37.43

S4.2 Reverse-trace reconstruction

The trace grid crosses fields $\{1, 8, 32, 128\}$, versions $\{1, 10, 100\}$, and total records $\{1, 100, 1000\}$, giving 36 cells and 14,400 timed observations across two arms. Valid unrelated records are seeded before timing. The point-access arm enumerates identifiers and performs real point loads for

the certificate, certificate–evidence, and transition repositories. Both arms run the same current verifier. Equality of the complete trace object is checked after each timed call.

Table S9: Complete current trace grid. Latency is p50 / p95 in milliseconds; each arm has 200 timed observations after five warmups. SQL counts are constant within each arm/cell.

Fields	Versions	Records	Bulk access	Point access	Bulk SQL	Point SQL
1	1	1	6.06 / 8.38	7.17 / 10.58	12	15
1	1	100	6.15 / 9.27	7.33 / 10.88	12	15
1	1	1000	7.27 / 11.24	8.71 / 13.25	12	15
1	10	1	8.12 / 14.57	22.44 / 35.33	12	42
1	10	100	7.45 / 12.12	20.65 / 32.48	12	42
1	10	1000	7.85 / 12.69	21.72 / 34.71	12	42
1	100	1	20.52 / 24.14	152.07 / 163.52	12	312
1	100	100	22.35 / 36.98	164.96 / 205.99	12	312
1	100	1000	24.59 / 38.62	177.42 / 219.39	12	312
8	1	1	7.49 / 10.93	18.32 / 27.82	12	36
8	1	100	8.12 / 14.26	19.67 / 35.65	12	36
8	1	1000	7.86 / 12.32	19.76 / 29.46	12	36
8	10	1	10.99 / 16.66	37.17 / 52.74	12	63
8	10	100	9.05 / 12.07	31.33 / 42.47	12	63
8	10	1000	11.00 / 15.05	37.47 / 48.33	12	63
8	100	1	28.09 / 37.51	169.89 / 229.18	12	333
8	100	100	29.48 / 38.74	178.35 / 206.82	12	333
8	100	1000	32.39 / 48.33	196.73 / 247.61	12	333
32	1	1	10.95 / 15.00	52.95 / 66.65	12	108
32	1	100	12.95 / 19.41	61.33 / 77.08	12	108
32	1	1000	11.48 / 14.61	54.53 / 73.08	12	108
32	10	1	16.16 / 22.78	77.03 / 98.56	12	135
32	10	100	14.55 / 18.50	67.70 / 80.59	12	135
32	10	1000	14.55 / 19.94	68.61 / 85.06	12	135
32	100	1	56.13 / 70.49	245.75 / 298.96	12	405
32	100	100	56.33 / 76.65	247.93 / 299.68	12	405
32	100	1000	55.65 / 74.35	241.16 / 303.78	12	405
128	1	1	30.94 / 43.03	221.63 / 290.86	12	396
128	1	100	28.20 / 35.46	199.90 / 256.49	12	396
128	1	1000	28.68 / 43.26	209.81 / 256.03	12	396
128	10	1	42.55 / 61.87	245.20 / 304.18	12	423
128	10	100	38.42 / 48.50	221.52 / 255.50	12	423
128	10	1000	39.69 / 48.16	233.82 / 285.43	12	423
128	100	1	155.23 / 206.20	476.07 / 581.57	12	693
128	100	100	141.00 / 176.70	436.33 / 489.88	12	693
128	100	1000	170.24 / 216.75	509.09 / 595.85	12	693

At 128 fields, 100 versions, and 1,000 total records, the medians are 170.24 ms for bulk access and 509.09 ms for point access, a point/bulk ratio of 2.99. Statement counts are 12 and 693. The record count varies background population for trace access; it does not measure a cross-record atomic transaction. This comparison is not a rerun of the historical preoptimization algorithm and is not combined with historical-machine timings. The archive retains paired differences and bootstrap intervals; those intervals describe repeated observations on this machine, not cross-deployment variation.

S4.3 Storage-population boundary

At 1,000, 10,000, and 100,000 transitions, the full-schema database sizes are 1.52, 13.16, and 134.53 MiB; simplified storage uses 0.48, 2.88, and 28.45 MiB. Incremental bytes per transition are 1,085.4, 1,077.2, and 1,112.3. MiB uses 2^{20} bytes. Rows are directly populated using a synthetic single-field history with stable evidence references, then checkpointed. They are not evidence that every populated row passed admission. The simplified schema has fewer audit obligations and is not a functionally equivalent alternative. External evidence and raw model-output storage are excluded.

S5 Historical agent integration and annotation

S5.1 Distinct evidence and denominators

The historical restricted-agent benchmark has 1,260 planned executions across three requested configurations, each with 420 runs. The configuration mapping is G1 = `gpt-5.6-luna`, G2 = `gpt-5.6-terra`, and D1 = `deepseek-v4-flash`; these archived requested names identify recorded requests, not immutable model weights or new calls made for this revision. A prior pinned-plugin fixture has ten passing integration cases, and six AI-origin lifecycle cases match their declared outcomes. These support exercised interface behavior rather than current provider comparisons.

Among 360 planned benign runs, 320 satisfy the frozen strict trajectory and 331 the post-hoc terminal-endpoint rule. Twenty-five lack a scorable behavior verdict, leaving 335 evaluable runs. The eleven-case difference consists of extra proposal or parent-verification calls that violate the trajectory criterion while reaching the terminal result. It measures sensitivity to the scoring construct. The strict yield is 320/360 (88.89%); conditional strict completion is 320/335 (95.52%). The endpoint counterparts are 331/360 (91.94%) and 331/335 (98.81%). Missing cases are retained, not silently excluded from planned-run yields.

Across all scenarios there are 93 recorded runtime failures: 76/420 for D1, 4/420 for G1, and 13/420 for G2. Some failures retain an authority verdict, and three harness failures retain a behavior verdict; evaluability follows available evidence rather than success status alone.

Separately, no unauthorized mutation is recorded in 720 fixed invalid-tuple admission calls or 179 capability-unavailable branches. The latter do not call admission. Their sum is accounting, not 899 independently sampled attacks. Zero violations in this constructed workload do not establish a population failure bound or general prompt-injection security. The restricted interface leaves admission with the trusted host.

S5.2 Human annotation and unresolved disagreements

One author (X.W.) and one non-author volunteer coded benign terminal completion. They agree on 359/360 original labels (99.72%; three-category Cohen's $\kappa = 0.974$). Both label 331/335 behavior-evaluable cases as terminally complete. L.H. separately reviews the eleven strict-negative/endpoint-positive cases and retains their strict-negative classification; this is not a replacement of the original completion labels. The separate T153 inter-human discrepancy is adjudicated in the historical record. Author involvement and residual textual cues limit independence and blinding.

A different textual-acknowledgment audit has 325 outputs labeled by X.W. Agreement with repaired rule v2.1 is 286/325 (88.00%; $\kappa = 0.644$). A3 contributes 32 of 39 disagreements and lacks a uniquely specified target field. Those 39 cases have no completed human adjudication; later unblinded AI proposals do not replace their labels. This audit measures human–rule agreement, not inter-human reliability or factual correctness. Complete labels, rationales, confusion matrices,

Table S10: Benign B1–B4 strict-trajectory completion by configuration. Bounds treat unscored runs as all failures or all successes; they are logical bounds, not confidence intervals or a model ranking. Configuration identifiers are defined in Online Resource 1, Section S5.

Config.	Planned	Unscored	Strict/evaluable	Rate	Bound
G1	120	1	114/119	95.80%	95.00–95.83%
G2	120	12	105/108	97.22%	87.50–97.50%
D1	120	12	101/108	93.52%	84.17–94.17%
All	360	25	320/335	95.52%	88.89–95.83%

Table S11: Runtime-failure terminal classes crossed with endpoint evaluability.

Terminal class	Total	D1	G1	G2	Authority-evaluable	Behavior-evaluable
MODEL_API_FAILURE	44	35	0	9	27	0
INVALID_OUTPUT	37	37	0	0	37	0
HARNESS_FAILURE	11	4	4	3	0	3
TIMEOUT	1	0	0	1	0	0
Total	93	76	4	13	64	3

and resampling sensitivity remain in the pinned historical archive. Neither annotation exercise is part of E2's scripted-browser denominator.

S6 Additional protocol and scoring details

This section records details beyond the main protocol. Online Resource 1, Section S1 gives the complete E1/E2 input and case lists, environment, execution commands, rejection-audit baseline, and source receipts. S4 gives all timing and storage cells; S5 gives historical integration endpoints. Those records retain their original denominators.

S6.1 Comparator and query observations

The event journal is a separately written `sqlite3` comparator developed for this study, rather than an externally deployed system or an independent-team replication. Both configurations retain immutable candidates and events, review context, version checks, transactional updates, and complete field sources. Context-level authorization retains field, version, evidence, producer, and proposal context while excluding certificate identifier, internal candidate identifier, and instance creation time. Exact authorization adds the reviewed-candidate identity. The reference arm invokes the actual transactional admission service.

The instance policy requires the authorized candidate instance. The value/context policy permits another instance only when the retained observations agree. Evidence substitutions change constructed metadata rather than image contents. The archived inputs are deterministically selected without using task-success labels. Their twelve configuration–scenario cells contain four distinct values, and three synthetic controls add the type boundaries listed in S1.

An independent reader scores six queries for each of three fields. Answers are correct, wrong, ambiguous, or unavailable. Rejection requires the expected constraint violation or injected interruption as well as an unchanged logical fact-database digest; an arbitrary exception is not a successful rejection. The baseline follows authorization preparation, so recorded review intent can predate a rejected admission. The service-generated reference bundle is captured before submission to the transaction. The strengthened-baseline continuity rerun remains outside the 495-execution denominator.

S6.2 Browser fixture

An independent Playwright driver reads the displayed candidate identifier, proposal, and authorized value from the DOM before the authorization click. The original `ReviewForms.confirm` service and experimental server-held session gate process the same declared interactions. Six initial confirmations prepare replay cases and are setup operations beyond the sixty scored cases. The decimal input avoids JavaScript normalization of an integral float. The fixture uses fixed trusted principal labels; its session database is separate from the fact database.

S6.3 Cost measurement boundaries

Each timing cell uses five warmups and 200 observations per arm, randomly interleaved with seed 20260923. Warmups, database copying, and adapter construction are excluded. The exact journal opens its connection during construction; the reference ORM can obtain its first connection during confirmation. SQL hooks also count trigger execution differently, so absolute cross-implementation statement totals are not compared.

For the separate manual-source materialization ablation, full admission generates a valid relational delta outside the reduced-arm timer. Applying that delta retains complete snapshots, source maps, authority rows, transactionality, and CAS. The resulting state must match the precomputed relational fingerprint. Random review identifiers and timestamps prevent a byte-identical database claim across separate full admissions. This ablation and the recognition-source mechanism comparison use different workloads.

The point-read trace arm changes only the certificate, certificate–evidence, and transition repository reads. Valid unrelated forms are seeded once and copied outside timing. The complete

returned trace object must equal its counterpart after every invocation. Storage fixtures use single-field version growth and stable evidence references, with WAL/NORMAL and checkpointing before footprint measurement. Direct schema population measures storage, rather than admission of every populated row.

S6.4 Historical scoring

The strict benign criterion requires the specified exact call sequence; its stale scenario also requires the word `stale`. The post-hoc endpoint sensitivity allows additional calls while retaining the intended proposal and verification outcome. Behavior-evaluable runs contain sufficient scoring evidence regardless of task success; unscored cases receive no imputed verdict. Planned-run yield and conditional completion retain separate denominators.

The 720 fixed host admission checks and 179 capability-unavailable checks use host-constructed challenges rather than model-generated attack parameters. The latter branch makes no admission call. Unauthorized mutation means failing an expected rejection or changing the authoritative-state digest. Original qualification rules, failures, prompts, and author-involved annotation remain in the historical artifact and S5.

S7 Extended validity details

The main discussion states the approval-policy choice, trusted review boundary, formal scope, and cost interpretation. This section preserves additional boundaries for the archived evidence.

S7.1 Observation and implementation scope

An omitted observation class cannot be reconstructed by decoding an opaque candidate handle. Other observation models may combine or derive distinctions differently. The paired constructions establish conditional class-wise irredundancy, rather than universal minimality or completeness over every admission failure. Several Alloy assertions check immediate consequences of definitions; selected projection instances and mutants test the abstraction interface, rather than a refinement theorem over every execution.

Canonical JSON equality governs change detection, authorized and committed values, and tracing. The repaired integer/float and numeric/Boolean cases show why these checks need the same value-equivalence relation. Targeted regressions cover selected boundaries, rather than every serialization or migration behavior. The concrete prefix trace exceeds the formal single-step anchor check in some respects; same-value admission and schema uniqueness also differ as stated in Section S2.

Evidence checks compare stored digests and canonical locators without re-reading external bytes. Replacing external evidence while retaining its metadata therefore needs immutable storage or another integrity check. Authentication, serialization, policy state, hashing, and transaction execution remain trusted. A revocation visible before the commit-entry policy check causes rejection; a later revocation does not cancel that in-flight transaction. Deletion, redaction, tombstones, retention expiry, distributed revocation, and other database engines require further design and evidence.

S7.2 Constructed and historical studies

The 495 E1 executions are factorial constructed cases using seven distinct proposal values, rather than independent applications. The comparator is a same-study implementation and evidence contexts are constructed metadata. E2 is an instrumented local fixture, rather than the deployed product interface or a human-subject study. Its counts describe exercised interactions; authentication, attention, hostile-host resistance, and crash atomicity across the session and fact

databases remain unevaluated. The session gate is an experimental extension, not a property attributed to the original path.

Historical hosted-agent runs have separate scoring criteria, missing runtime outcomes, fixed host checks, and author-involved annotation. Fixed challenge tuples measure the exercised integration checks rather than autonomous attack efficacy or population failure probabilities. E1–E3 make no hosted-model calls.

S7.3 Measurement interpretation

Admission timings include differences in schema, ORM, certificate work, application features, and connection lifecycle. The prevalidated-materialization arm moves validation and planning outside its timer and cannot replace full admission safely. The trace experiment changes access within the same current verifier. Twelve SQL statements across the grid do not imply constant total cost: transferred rows, indexes built in memory, and predecessor traversal still grow with history and field count.

One-machine interleaving limits order effects while retaining scheduling, caching, and connection differences. Repetition-based intervals describe that run, rather than uncertainty across machines or deployments. Storage excludes external evidence bytes, model-output files, and production retention policies; the reduced schema has fewer audit obligations. An independent application with operational reviewers, records, and policies would provide the next test of transferability.

S8 Prospective policy experiment: design and scoring

The prospective study is frozen under `phase2-formal-v1.0-9fd97d11ceaed652`. Its master SHA-256 is

```
7773563b40c8db30d6def121915f50fbad032537edde32afd4ba39ec3eea744b
```

The local freeze timestamp is 2026-10-06T18:55:47.225978+00:00. This is a local prospective freeze, not an externally registered study. The 16-arm online pilot, availability probes and deterministic fixtures are separate engineering records and contribute no arms to formal effect analysis.

S8.1 Task and assignment manifests

The task manifest contains 128 rendered task instances, distributed L32/N32/G16/E16/V16/R16. Each is evaluated with configurations A and B under context and bound policies, yielding 256 pairs and 512 planned arms, repetition one. The assignment manifest fixes each pair, its scenario/configuration and adjacent policy order. Order is balanced within scenario/configuration using the frozen seed. Two provider lanes each execute one pair at a time.

Parameter vectors expose finite variation in business template, correction direction/magnitude, proposal/authorized relation, evidence layout and locator, source configuration, predecessor and history depth, retained metadata, reference-note content/source, candidate envelope, evidence snapshot organization, field ordering, correction location and task wording. Rendered evidence and authoritative predecessor history are stored in the task inputs. Input hashes refer to the actual rendered content. A unique vector is not treated as an independent production workflow or a new scenario family.

S8.2 Shared prefix and common interface

Each pair has one actually sampled prospective prefix, ending at the first valid authorization. The visible messages and persisted checkpoint state are copied to the two arms. Subsequent

provider responses are independently obtained under identical tool definitions and prompts. The study copies observable history and host state, not inaccessible provider hidden state. Both policies retain the reviewed instance identifier.

The common serialized action facade presents `get_state`, `inspect_evidence`, `propose`, `verify`, `request_authorization`, `commit` and `get_receipt`. The model chooses one action; the host actually executes it and inserts the result into the next visible conversation. This facade is shared across providers and is not described as provider-native function calling. The full prompt and tool-schema manifests are preserved in the frozen artifact.

S8.3 Scenario-specific semantics

L provides legal corrected execution and unchanged-field source preservation. N requires a refreshed preview after review but allows the agent to reuse the original reviewed candidate, request a fresh authorization or otherwise select a continuation; generating a preview does not itself mandate a mismatched commit.

G creates a context-equivalent new candidate and performs one deterministic index commit with the prior grant. The origin is `harness_handoff`. Context can accept it and bound can reject it under the disclosed instance rule. The actual result is then included in subsequent sampled agent decisions. The index action is excluded from agent tool-call counts and is not an autonomous agent attack.

E publishes E2 with changed bytes, hash and locator, keeping the proposal value equal. E1 remains available for audit but is superseded for new review/execution by the common current-evidence guard. Reusing E1 with the old grant can therefore yield `EVIDENCE_SUPERSEDED`; substituting an E2-based candidate under the E1 review yields context mismatch with same value and unequal context. Fresh E2 review provides legal recovery under both policies. This exposure affects the shared evidence guard, not the policy's unique instance conjunct.

V externally advances authoritative version and reference-note source. The goal thereafter preserves that external note/source. Refreshing state, constructing a current-predecessor candidate and obtaining authorization enables a fresh commit; using the old candidate fails freshness under both policies.

R returns `CAPABILITY_UNAVAILABLE` on the first receipt access after a successful commit while keeping authoritative state readable. The stored receipt remains durable. Identical arguments with the same idempotency key can return the prior success without a new transition. A different key under the consumed grant is semantic replay and is rejected. Confirmation via state or later receipt is recorded separately from additional execution.

S8.4 Independent endpoints and event-order taxonomy

Utility requires state goal attainment, required workflow steps and accurate final report. Integrity checks covered authoritative transitions for valid authorization, freshness, evidence/context integrity, authorized-value equality, correct sources/provenance, unchanged-source copy-forward and instance-compatible review-to-execution continuity. Proved violations dominate missing evidence. Integrity unknown is not replaced by success or failure, and a runtime error does not automatically imply $I = 0$.

Run-level continuity-failure incidence is one when at least one authoritative transition violates the instance requirement; it is zero only with sufficient transition coverage and no such violation. Executed-substitution fraction uses transitions as its denominator and is a separate diagnostic. Exact idempotent success receipts create no new transition.

Delivery is proved by a received next provider response tied to the feedback-containing input. Legal recovery paths are computed from the recorded rejection-time state and capabilities. Available paths can include valid reviewed reuse, review of a current candidate, fresh-state

proposal/review, current-evidence proposal/review and state/receipt confirmation. Future success is not used to decide whether the earlier event was recoverable.

The frozen event-order classifier records reuse-reviewed, reverify, reproposal, reauthorization, state-refresh, evidence-refresh, stale-reuse, repeated-invalid, fresh-id-invalid-loop, abandon, timeout, runtime-failure, hallucinated-success and successful confirmation via state/receipt. Successful reauthorization requires an actual successful authorization result; a failed attempt is not relabeled as a successful recovery. Repeated invalidity counts rejected subsequent commits, preserving candidate identities. Hallucinated success requires a known contradiction of the final report; unknown state alone does not establish it. Multiple labels and component actions remain visible even when a deterministic priority selects one primary path. No language model freely classifies trajectory text.

S8.5 Analysis denominators and uncertainty

All 512 arms remain in the planned ledger. A failed shared prefix gives both arms unsuccessful utility; audit evidence separately determines integrity. The ledger distinguishes unstarted arms during a running/paused batch from terminal failures. Final effect inference requires a complete collection status.

Aggregate effects average the two configuration-specific paired deltas within each task, then average over 128 tasks. Scenario-stratified task-cluster resampling retains both configurations and both policy arms, including associated events. The frozen bootstrap has 10,000 iterations, seed 20261007 and percentile 95% intervals. These intervals summarize controlled task variation. Exact two-sided McNemar tests are secondary and configuration-specific. Unknown continuity outcomes retain conservative identification bounds; neither unknown nor provider failure is silently removed.

Conditional recovery summaries are post-treatment descriptive results. G-bound eligibility follows checkpoint coverage; the 32 planned G-bound arms and eligible/actually delivered counts are separate. First-episode and any-episode denominators remain distinct, and overlapping episodes are not independent samples. All U=0/I=0/unknown arms appear in the failure ledger before examples are selected.

Recovery efficiency retains separate tool calls, model turns, authorization attempts, proposals, verifications, elapsed time and provider usage. Budget-capped time-to-success is elapsed suffix time at an accurate completed report, capped at 720 seconds; unsuccessful arms receive that cap and preserve their actual terminal time separately. Prefix latency belongs to the logical trajectory, while billed prefix usage is counted once per pair. Unobserved token components and currency cost remain null.

S9 Deployment, collection and recovery records

Configuration A uses requested alias `gpt-5.6-terra`, Codex CLI `0.160.1`, low reasoning, the account-backed route and the frozen existing proxy. The transport does not expose an actual returned model identifier or immutable revision. Configuration B uses requested alias `deepseek-v4-flash` at the recorded compatible endpoint; the precollection probe returns that alias, temperature is zero and the output cap is 4096. Neither returned alias certifies an immutable snapshot. Top-p and seed were not explicitly supplied; no default is invented. Credentials are outside the repository and are never included in requests archived for analysis.

The shared resource-policy family caps the prefix at twelve tools/eight responses/300 seconds and each suffix at twenty-four tools/twenty responses/720 seconds. Requests have a 90-second timeout with at most two outer transport retries, waits of 2/8 seconds extended by Retry-After subject to remaining run time. A has no supported upstream output-token cap; B has its recorded 4096 cap. Currency cost is unobserved. Native CLI reconnection is recorded separately from the

outer attempts. A logical history can receive one generic malformed-action feedback; it receives no task-specific repair hints.

Transient HTTP 429/5xx, timeout, URL/reset/abort and mid-read failures retain attempt records and may consume the frozen retries. Credential/balance/model unavailability and malformed provider envelopes terminate as recorded runtime failures. Semantic rejection is tool feedback, not a reason to regenerate a task. Receipt-path unavailability belongs to R's controlled capability exposure rather than transport retry.

Atomic safe-boundary files retain messages, state, pending feedback, response/tool counters and received responses. Resume reuses a cached received response and skips every terminal pair, including failures. Downtime counts in elapsed budgets. An interrupted semantic step preserves actual durable state, is not reexecuted and is not rolled back to an old boundary. Missing initialized-arm boundaries prevent empty-history resampling. A process-held lock prevents a second coordinator and releases on process death. Observable deployment changes stop both lanes at subsequent decision boundaries; frozen scientific-input changes require a retained old batch and a new version/freeze.

The first collection was paused after five terminal pairs by a Windows safe-boundary replacement PermissionError. Those ten arms, including the affected partial/unknown arm and the other lane's recorded stop, remain in the original denominator. Inspection found the original coordinator terminal; 250 subsequent atomic-write probes succeeded, and the same frozen code resumed unstarted pairs without resampling any terminal pair. Runtime logs and the pause audit are retained alongside the batch. The complete runtime ledger retains this interruption and all affected arms; the operating-system handle owner was not established.

S9.1 Artifact contents and exact reproduction

The prospective artifact directory is `research/agent-policy-phase2-formal-2026-10-07`. Its `frozen-formal/` directory retains the final protocol, rendered task and assignment manifests, deployment/prompt manifests, tool and trajectory schemas, scorer definition, analysis plan, failure matrix, input and source hashes, environment and master manifest. The accompanying source contains the actual runner, independent oracle, event classifier, exporter and figure generator. All 128 deterministic fixtures and 57 runner/scorer tests were checked before collection. These engineering records are excluded from formal online statistics.

Offline verification. Extract the hash-verified handoff archive into a new directory, preserve that extraction, and work on a copy. From the extraction root, run the original analyzer with a fresh output directory:

```
python research/agent-policy-phase2-formal-2026-10-07/analyze_formal.py \
  --output fresh-analysis --figures
```

The backslash continues one shell command onto the next line. This analysis uses archived responses and persisted state and makes no new model calls. Install the archived requirements in an isolated environment when reproducing its dependency versions. Compare the six CSV tables and the parsed analysis summary semantically with the deposited export before using regenerated figures.

In the manuscript directory, `python verify_evidence.py` checks the counts, baseline/controlled hashes, historical chains and archived formal receipts. The present verifier normalizes Windows separators in the source manifest for Linux portability; it changes no source hash or scientific check. From `research/authorization-granularity-phase1-2026-10-06/formal/`, run `python -m unittest discover -s tests -q` for the executable witness tests. Fresh online collection is a separate operation requiring the frozen deployment, provider access and the cohort-specific collection rules; it is not part of these reproduction commands.

The final analyzer produces scenario/configuration/policy outcomes, per-transition continuity, delivered-feedback recovery, per-arm operational components, paired primary effects, all-arm failure archaeology and unique-prefix usage. Formal recovery/effect figures require complete data. N/G flow and paired component plots are supplied as vector and high-resolution outputs in the result package.

S10 Positioning and reconstruction traceability

The closest execution-oriented comparisons are the posted preprint versions of Continuity Kernel (He and Yu, `https://arxiv.org/abs/2608.11632`) and commit-time authorization (Santos-Grueiro, `https://arxiv.org/abs/2607.10487`). Their respective units are an accepted branch-head transition and a durable effect licensed by a current authority witness. The present comparison fixes a corrected authorized value and retained context while varying reviewed-instance equality, then traces actual rejection feedback through paired agent continuations. This distinguishes the evaluated question without claiming that those frameworks cannot represent corrections. MemTxn (`https://arxiv.org/abs/2607.27834`) and MutMem (`https://arxiv.org/abs/2608.02843`) provide further memory-governance context. The records accessed on 7 October 2026 do not display linked journal references; this comparison refers to the posted preprint versions. Exact binding, approval, provenance and generic recovery are acknowledged foundations, rather than claimed inventions.

The reconstruction blueprint, old-content disposition and claim-evidence audit are editorial records outside the submitted main narrative. They prevent historical persisted corrections, deterministic fixtures or pilot arms from being reported as formal task success. The delivered verification report maps the main-text numerical claims to the immutable controlled archive, Phase 1 audit or complete prospective result table.

S11 Complete prospective outcomes and failure archaeology

Collection began at 2026-10-06T19:05:53 UTC and the final coordinator record is dated 2026-10-06T21:50:05.535328 UTC (7 October, Asia/Shanghai). All 256 pairs are terminal, accounting for the entire 512-arm planned ledger. A terminal ledger entry does not imply a complete model trajectory: the original interrupted N pair retains a partial context arm and an unstarted bound arm. Both have unsuccessful utility and unknown integrity. The final exporter recomputes scores from persisted events and states, rather than relying on cached coordinator scores. The final analysis status is `formal_inference_allowed=True`.

S11.1 Original aggregate, recovery and cost tables

The following tables preserve the original aggregate outcomes and conditional recovery and operational summaries under their original denominators. Checkpoint counts use reached-review events and matching saved-state hashes. All 128 B pairs have these records, although only 127 pair-summary flags survive: N-020-B has a partial context suffix and an unstarted bound suffix. Its two integrity scores remain unknown.

S11.2 Complete scenario/configuration/policy table

Table S15 gives the full original denominator. Checkpoint counts refer to the shared prospective prefix. Runtime denotes the frozen exporter's prefix/runtime/audit-state failure classification; malformed model output is a separate terminal class. Counts in the rejection column are arms with at least one rejected admission, not numbers of all rejection episodes. Receipt unavailability is separately classified feedback and is not an admission rejection.

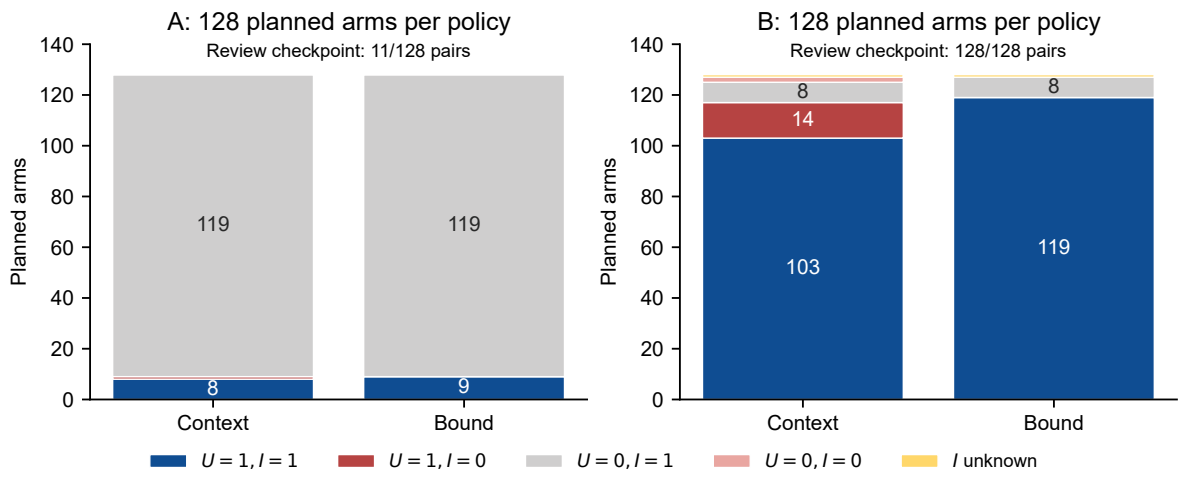


Figure S2: Original planned-ledger utility and integrity. Each policy/deployment column contains 128 arms. The large $U = 0, I = 1$ component in A reflects covered state with provider failures, predominantly before review. Small joint cells and both unknowns remain explicit in the main joint-outcome table and the complete table below.

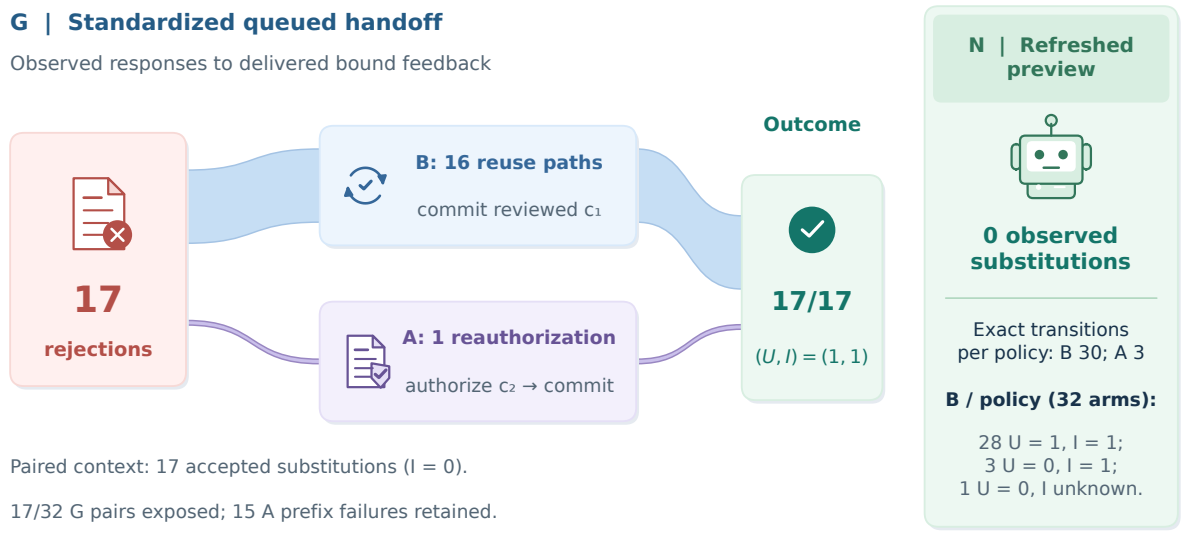


Figure S3: Original observed paths after standardized mismatch feedback. All 17 exposed G-bound continuations recover with $(U, I) = (1, 1)$: B reuses the reviewed candidate in 16 cases, and A reauthorizes the handoff candidate in one. The separate N panel retains agent-managed preview outcomes and unknown coverage. The figure reports observed paths, not the frequency of naturally occurring substitutions.

Table S12: Original complete planned-ledger outcomes. Every row contains 128 arms. A covered run without an unauthorized transition can have $I = 1$ despite unsuccessful execution. Unknowns are the two N arms affected by the initial infrastructure interruption. Supplementary A outcomes retain their own denominator.

Deployment	Policy	$U = 1$ $I = 1$	$U = 1$ $I = 0$	$U = 0$ $I = 1$	$U = 0$ $I = 0$	I unknown	$U = 1$	Checkpoint pairs
A	Context	8	1	119	0	0	9	11/128
A	Bound	9	0	119	0	0	9	11/128
B	Context	103	14	8	2	1	117	128/128
B	Bound	119	0	8	0	1	119	128/128

Table S13: Delivered-feedback continuations in original B. Each row's denominator is delivered episodes, not all planned tasks. All listed episodes have a legal path at rejection time. E, V and R involve shared checks/capabilities; only G exposes the policies' unique instance rule. Recovery requires a recorded recovery endpoint, $U = 1$ and $I = 1$.

Family	Policy	Delivered	Recovered	First meaningful action	Observed path
G	Bound	16	16	Commit (16)	Reuse reviewed candidate (16)
E	Context	16	15	Inspect evidence (15), propose (1)	Fresh proposal and authorization (15)
E	Bound	16	15	Inspect evidence (16)	Fresh proposal and authorization (15)
V	Context	13	12	Propose (7), state read (5), none (1)	Fresh proposal and authorization (12)
V	Bound	13	12	State read (10), propose (2), none (1)	Fresh proposal and authorization (12)
R	Context	9	9	State read (9)	State confirmation (9)
R	Bound	10	9	State read (9), none (1)	State confirmation (9)

Table S15: Complete formal outcomes by family, deployment and policy. C=context, B=bound; CP=checkpoint reached; U=task completion; Rj=arms with rejected admission; RT=runtime failures.

Family	Config	Policy	Planned	CP	U	$I = 1$	$I = 0$	Unknown	Rj	RT
E	A	B	16	2	1	16	0	0	1	15
E	A	C	16	2	1	16	0	0	2	15
E	B	B	16	16	15	16	0	0	16	0
E	B	C	16	16	15	16	0	0	16	0
G	A	B	16	1	1	16	0	0	1	15
G	A	C	16	1	1	15	1	0	0	15
G	B	B	16	16	16	16	0	0	16	0
G	B	C	16	16	14	0	16	0	0	0
L	A	B	32	3	3	32	0	0	0	29
L	A	C	32	3	3	32	0	0	0	29
L	B	B	32	32	32	32	0	0	0	0
L	B	C	32	32	32	32	0	0	0	0
N	A	B	32	3	3	32	0	0	0	29
N	A	C	32	3	3	32	0	0	0	29
N	B	B	32	32	28	31	0	1	0	0

Family	Config	Policy	Planned	CP	U	$I = 1$	$I = 0$	Unknown	Rj	RT
N	B	C	32	32	28	31	0	1	0	0
R	A	B	16	2	1	16	0	0	0	15
R	A	C	16	2	1	16	0	0	0	15
R	B	B	16	16	16	16	0	0	0	0
R	B	C	16	16	16	16	0	0	0	0
V	A	B	16	0	0	16	0	0	0	16
V	A	C	16	0	0	16	0	0	0	16
V	B	B	16	16	12	16	0	0	13	0
V	B	C	16	16	12	16	0	0	13	0

S11.3 Primary effects and unknown identification

The frozen aggregate completion estimator averages the two configuration-specific paired differences within each task. Bound completes 128/256 planned arms and context 126/256, with 256 planned arms per policy (512 across both policies). This is planned-ledger completion, retaining pre-review quota failures; it is not completion conditional on reaching review. The effect is 0.0078125, with task-cluster percentile interval $[-0.01953125, 0.03515625]$. Configuration A has 9/128 completions under both policies, and B has 119/128 bound versus 117/128 context. B's paired effect is 0.015625 with interval $[-0.0390625, 0.0703125]$. Its discordant pairs are five context-only and seven bound-only completions; the secondary exact two-sided McNemar p is 0.7744140625. A has no completion-discordant pairs.

Continuity is unknown in the interrupted N pair. The aggregate bound-minus-context failure-incidence difference has no identified single point estimate: lower and upper identified endpoints are -0.0703125 and -0.0625. Their bootstrap intervals are $[-0.08203125, -0.0625]$ and $[-0.07421875, -0.05078125]$. The intervals apply to their respective endpoints; they are not a conventional interval around an imputed estimate. In A, the observed planned-ledger difference is -0.0078125, interval $[-0.0234375, 0]$. In B, the one unknown per policy requires identification bounds. All intervals use the frozen 10,000 scenario-stratified task-cluster draws, seed 20261007.

The transition table has 261 observed authoritative transitions. All seventeen substitutions occur in G-context (A:1, B:16), pass the context policy and fail the instance oracle. These transitions satisfy the remaining evaluated guard, source and copy-forward relations. B's N arms contain thirty exact-instance transitions per policy; A's contain three per policy. A missing transition is not inferred to be a substitution.

S11.4 Feedback, absent categories and denominators

The complete delivered-feedback table contains 99 episodes: 93 in B and six in A. All are recoverable at their recorded feedback states. B contributes G-bound16; E-context16/E-bound16; V-context13/V-bound13; R-context9/R-bound10. A contributes G-bound1; E-context2/E-bound1; R-context1/R-bound1. No delivered episode occurs in L or N. For these data each feedback-bearing arm contributes one episode, but the analysis definition permits multiple episodes and does not treat them as independent samples.

G-bound recovery succeeds in all seventeen delivered episodes. B reuses the reviewed candidate in sixteen cases; A obtains new authorization in one. E in B recovers fifteen of sixteen per policy through fresh proposal/review; V recovers twelve of thirteen per policy. R's state-confirmation endpoint occurs in nine B episodes per policy. One additional B-bound R episode ends with correct run-level completion but no recorded recovery endpoint. A has one recovered E episode per policy, with a second context E episode interrupted during recovery; its context R state-confirmation action is followed by runtime failure before accurate task completion. Thus successful intermediate repair, episode recovery and terminal task utility remain different observations.

Table S14: B's paired operational components (bound minus context). "All" uses 127 pairs with both suffix records; G uses all 16 pairs. Elapsed time includes unsuccessful suffixes. Median and mean time answer different questions.

Component	All	G
Mean extra agent calls	0.071	1.000
Mean extra model turns	0.039	0.938
Mean extra authorizations	0.000	0.000
Median extra suffix seconds	0.69	14.54
Mean extra suffix seconds	1.29	8.84

Observed component labels comprise reuse-reviewed, reproposal, reauthorization, evidence-refresh, state-refresh and successful-confirmation-via-state, plus runtime-failure in two delivered episodes. The delivered table has zero reverify, stale-reuse, repeated-invalid, fresh-id-invalid-loop, abandon, timeout, hallucinated-success or successful-confirmation-via-receipt labels. Zero means absent under the frozen observable classifier, not that the corresponding failure is impossible. The original component arrays are preserved alongside primary path labels.

S11.5 Complete failure partition and provider coverage

The failure ledger has 273 distinct arms: 238 runtime-failed A arms; fifteen successfully completed context arms with instance violation; two unsuccessfully completed context arms with instance violation; sixteen other unsuccessfully completed B arms with covered intact integrity; and the two unknown N arms. These disjoint classes sum to 273. B's eighteen malformed-output terminations comprise the two unsuccessful violating G-context arms and the sixteen intact-integrity failures. No proved violation occurs outside G-context.

A's 117 failed shared prefixes yield 234 failed suffix-ledger entries. Four more suffix runtime failures occur in two reached-checkpoint pairs. The raw CLI stdout archives contain 357 provider error events identifying usage-limit exhaustion across the recorded attempts. This confirms provider quota failure rather than a semantic gate rejection. The entry-time availability probe succeeded; availability during collection is an observed resource outcome. Terminal failures are not retried as new formal tasks after the quota resets. Original A reaches eleven checkpoints and one G index exposure, so its original recovery observations retain those denominators; the separate completed supplementary evaluation is reported in Section S12.

S11.6 Operational components and accounting

Across B's 127 pairs with two suffix records, mean bound-minus-context differences are 0.070866 agent calls, 0.039370 model turns, zero authorization requests and 1.285598 seconds actual suffix time (median 0.687 seconds). G's sixteen B pairs have a mean and median of one extra suffix call (two pairs have zero, twelve have one and two have two), 0.9375 extra turns on average, no extra authorization requests, mean 8.844563 seconds and median 14.539 seconds extra suffix time. In A's single exposed G pair, the extra calls are a review request and a commit; the other fifteen G pairs fail before exposure.

The capped time-to-success endpoint retains all 128 pairs per deployment. B's mean paired difference is -9.744703 seconds overall and -76.5275 seconds in G. Its G median is $+14.539$ seconds. The failed context reports receive 720 seconds, accounting for the negative mean despite positive median execution friction. The raw table retains actual elapsed and capped times separately.

The provider usage records are preserved in their native components. A distinguishes input, cached input and reasoning output; B distinguishes input, cache reads and cache creation. Unreported fields remain null. Prefix usage is charged once for each of 256 pairs, then each actual

suffix's usage is added separately. These observations do not supply a common currency cost. The artifact retains every native usage record rather than inventing a conversion or duplicating a prefix across arms.

S11.7 Result files and independent inspection

The result directory contains seven primary exports:

`table1-outcomes.csv`; `table2-continuity.csv`;
`table3-recovery.csv`; `table4-utility-friction.csv`;
`paired-effects.csv`; `failure-archaeology-all-arms.csv`;
`analysis-summary.json`.

Derived manuscript aggregates are a separate read-only layer. The replication manifest hashes the source freeze, raw results and published tables. Reanalysis checks the master/source hashes and recomputes the independent scores before producing tables. The artifact preserves the original interruption and resume launch audits, all provider responses and the partial N state; it does not fabricate missing termination or final-state files.

S12 Supplementary paired evaluation of deployment A

S12.1 Cohort, freeze and collection provenance

This supplementary evaluation extends deployment A's online continuation coverage while preserving the original 256-pair, 512-arm ledger. Its 117 pairs were selected because their original prefixes ended before review with zero model responses and a raw provider quota-exhaustion error. The selection rule does not use task utility, integrity or observed behavioral performance. The selected task distribution is L29/N29/G15/E14/V16/R14. Tasks, prompts, tools, schemas, policy order, requested deployment configuration, resource rules and scorer are inherited from the original freeze.

The supplementary freeze is `phase2-quota-recovery-v1-c41e052eb585`, created at 2026-10-07 00:15:31.836500 UTC. Its launch is recorded at 00:15:49.7034786 UTC and the first selected attempt starts at 00:15:55.478361 UTC. The final selected attempt ends at 09:46:49.308457 UTC; the coordinator records COMPLETE at 09:46:49.341704 UTC (17:46:49 Asia/Shanghai). All 117 planned pairs and 234 planned arms are terminal, and all 117 shared review checkpoints were reached. There are 117 retained pair-attempt directories; transport attempts remain inside their respective trajectories. The earlier 45-pair snapshot is superseded for supplementary reporting by this complete ledger and remains an archival collection snapshot.

The collection rule retains every attempt. Quota failure before any model response pauses collection and preserves the attempt; a later authorized continuation would use a new attempt directory. Quota failure after a response preserves the terminal trajectory and pauses collection. Ordinary terminal failures are retained and are not automatically repeated. Interrupted work follows the original safe-boundary and resource rules. The complete collection uses one A lane and makes no B calls.

Deployment A retains requested alias `gpt-5.6-terra`, Codex CLI `0.160.1`, low reasoning, the same account route and frozen proxy configuration. The independently rescored archive has 1,474 received model responses with that requested alias, but the transport exposes neither a returned model identifier nor an immutable revision. Requested configuration continuity therefore does not certify identical provider-internal state across collection periods. A runtime `DEPLOYMENT_CHANGED` record is discussed below as a failed deployment-identity check, not as evidence that an actual model revision changed.

S12.2 Complete supplementary outcomes

Table S16 retains every scheduled arm. Context completes 114/117 and bound 116/117. The context joint cells are $(U, I) = (1, 1) : 99$, $(1, 0) : 15$ and $(0, 1) : 3$; bound has $(1, 1) : 116$ and $(0, 1) : 1$. All other joint cells and unknown integrity counts are zero. All fifteen context integrity violations are executed G substitutions, each with successful task completion; all evaluated remaining guard and provenance relations hold.

Table S16: Complete supplementary A outcomes. C=context; B=bound; CP=shared review checkpoint reached; U=task completion; Rj=arms with at least one rejected admission; RT=runtime failures. Each policy row retains its planned denominator.

Family	Policy	Planned	CP	U	$I = 1$	$I = 0$	Unknown	Rj	RT
E	B	14	14	13	14	0	0	12	1
E	C	14	14	12	14	0	0	14	1
G	B	15	15	15	15	0	0	15	0
G	C	15	15	15	0	15	0	0	0
L	B	29	29	29	29	0	0	0	0
L	C	29	29	28	29	0	0	0	1
N	B	29	29	29	29	0	0	0	0
N	C	29	29	29	29	0	0	0	0
R	B	14	14	14	14	0	0	0	0
R	C	14	14	14	14	0	0	0	0
V	B	16	16	16	16	0	0	7	0
V	C	16	16	16	16	0	0	7	0

The supplementary paired completion difference is $2/117 = 0.0170940171$, or $+1.71$ percentage points. Its exploratory 95% percentile bootstrap interval is $[0, 0.0427350427]$ ($[0.00, 4.27]$ points). The procedure applies the original scenario-stratified task bootstrap to the 117 selected A-only pairs, retaining both policies, with 10,000 draws and seed 20261007. This application conditions on the selected cohort and is not the original frozen two-deployment estimator. There are two bound-only and zero context-only completions; the secondary exact McNemar $p = 0.50$.

The supplementary continuity-failure difference is $-15/117 = -0.1282051282$. Because every G pair has difference -1 and every other pair has difference zero, the fixed-scenario bootstrap returns the same continuity difference in every draw. This degenerate resampling interval reflects the observed within-family pattern and fixed family composition, not zero uncertainty across deployments, periods or broader tasks. There is no unknown-integrity identification interval in this cohort.

S12.3 Delivered feedback and recovery paths

The independent event-order classifier identifies 83 delivered recoverable feedback episodes. Every episode has a legal path at its recorded feedback state. Table S17 gives the observed actions and outcomes with episode-specific denominators. All fifteen standardized G-bound mismatch episodes end in accountable completion: eight first commit the reviewed candidate under the existing grant, and seven first request fresh authorization for the handoff candidate. The corresponding context index actions admit all fifteen substituted candidates. The N family contains twenty-nine authoritative transitions per policy; all are reviewed-instance compatible and both policies complete 29/29 tasks without delivered mismatch feedback.

Table S17: Supplementary A feedback episodes. Recovery requires a recorded recovery endpoint with $U = 1, I = 1$. Counts are conditional on delivered feedback and are not independent task samples.

Family	Policy	Delivered	Recovered	First meaningful action	Observed path
G	Bound	15	15	Commit (8); authorization request (7)	Reviewed-candidate reuse (8); reauthorization (7)
E	Context	14	12	Inspect evidence (13); final/no action (1)	Reauthorization (12); reproposal (1); abandonment (1)
E	Bound	12	12	Inspect evidence (12)	Fresh proposal and authorization (12)
V	Context	7	7	State read (7)	Fresh proposal and authorization (7)
V	Bound	7	7	State read (5); propose (2)	Fresh proposal and authorization (7)
R	Context	14	14	State read (14)	State confirmation (14)
R	Bound	14	13	State read (13); final/no action (1)	State confirmation (13); no observed endpoint (1)

One E-bound task completes without a delivered evidence rejection, so its 13/14 task-completion count exceeds the twelve episode recoveries. One R-bound task ends with $U = 1, I = 1$ without a recorded post-feedback recovery endpoint, preserving the distinction between run-level utility and the stricter episode endpoint. Raw component arrays, including evidence refresh, state refresh, reproposal, reauthorization and abandonment, accompany the primary path labels.

S12.4 Paired costs and complete failure records

Table S18 reports whole-suffix bound-minus-context differences, retaining unsuccessful suffixes. G's fifteen task pairs have mean 1.60 additional calls (exploratory 95% bootstrap interval $[0.93, 2.33]$), median one, mean additional actual time 17.9667 seconds (interval $[10.1562, 26.3020]$) and median 14.532 seconds. Recovery actions alone comprise one commit in each of eight reuse episodes and an authorization request plus commit in each of seven reauthorization episodes. Whole-suffix paired differences also reflect other continuation actions, so these two cost summaries need not coincide.

These intervals retain the archived seed 20261007 (10,000 draws). The secondary call-cost analysis in Online Resource 2, A12 uses seed 20261023 and gives $[0.87, 2.33]$ for the same ordered pairs and percentile estimator; the difference is Monte Carlo variation.

Table S18: Supplementary A paired operational components. "All" retains all 117 pairs; G retains its 15 pairs. Times are seconds. Capped time-to-success assigns 720 seconds to unsuccessful arms and is distinct from actual suffix time.

Component	All	G
Mean extra agent calls	0.222222	1.600000
Median extra agent calls	0	1
Mean extra model turns	0.230769	1.600000
Mean extra authorizations	0.076923	0.466667
Median extra actual suffix time	2.297	14.532
Mean extra actual suffix time	2.926761	17.966667
Median extra capped time-to-success	2.234	14.532
Mean extra capped time-to-success	−8.095017	17.966667

The complete failure-archaeology table contains nineteen distinct arms: fifteen successfully completed G-context arms with instance violations, three runtime terminations and one unsuc-

cessful agent-final report. All four unsuccessful arms retain covered integrity $I = 1$. FORMAL-L-010-A/context reaches the state goal but terminates with ACCOUNT_QUOTA_EXHAUSTED before completing the required report. FORMAL-E-015-A/context terminates on DEPLOYMENT_CHANGED; its paired bound entry retains the same code and the runner's prefix_failure terminal label despite the shared pair having reached review. FORMAL-E-009-A/context ends with the classifier's abandonment label. None is dropped from the supplementary denominator.

The DEPLOYMENT_CHANGED code records a deployment-identity check failure. For A the frozen check compares the observable Codex CLI version with the expected SDK string; no returned-model identifier is configured. The record therefore does not establish that the provider changed its model revision. Subsequent requests remain guarded by the same frozen identity check. Native transport records, received-response metadata and the attempt ledger preserve these distinctions. Unreported provider usage stays missing; it is not replaced with zero or converted into an invented currency cost. Shared-prefix usage is charged once per pair, and failed transport attempts remain separately inspectable.

S12.5 Source mapping, verification and offline reproduction

The raw supplementary directory is research/agent-policy-phase2-quota-recovery-2026-10-07/. Its manifest links each supplementary pair to the original pair, selection evidence, unchanged frozen inputs and retained original file hashes. Verification reproduces all 117 eligible pairs and confirms 15,832 original result files unchanged. The independent pass validates 7,989 events, makes 468 exact cached-score comparisons and records zero score-validation errors or violations. These checks concern the archived scoring and provenance obligations; they do not certify an unobserved provider revision.

The parent freeze master hash is

7773563b40c8db30d6def121915f50fbad032537edde32afd4ba39ec3eea744b.

The supplementary manifest SHA-256 is

3f11ebba8fb43b7ce2582e229678366a2338751505e61a667f02aea4c9b82e8f.

The complete raw archive is research/phase2-results-snapshots/2026-10-07-complete/quota-recovery-complete.zip, SHA-256

536f8b7d784d1af118145affd90c3d613b67ad44d9c5b3607b463bcebeacbec1.

The analysis adapter at research/phase2-followup-analysis-2026-10-07/analyze.py reads the complete supplementary records and invokes the unchanged frozen scorer. Its outputs/ directory contains table1-outcomes.csv, table2-continuity.csv, table3-recovery.csv, table4-utility-friction.csv, paired-effects.csv, failure-archaeology-all-arms.csv and analysis-summary.json. It also exports the attempt ledger, scored arms, source-file hashes, validation ledger, unique-prefix accounting and native usage records. These derived files remain separate from the immutable raw collection.

From the repository root, reproduce the supplementary analysis offline into a new directory within the analysis package:

```
python research/phase2-followup-analysis-2026-10-07/analyze.py
  --output research/phase2-followup-analysis-2026-10-07/fresh-output

python research/phase2-followup-analysis-2026-10-07/audit_usage.py
  --output research/phase2-followup-analysis-2026-10-07/fresh-output

python research/phase2-followup-analysis-2026-10-07/verify_outputs.py
  --output research/phase2-followup-analysis-2026-10-07/fresh-output
```

Run the three commands in order; each wraps over two lines. They recompute scores, audit native provider-attempt accounting and verify the derived outputs using retained records, with no model calls. The original verification entry point remains `research/agent-policy-phase 2-quota-recovery-2026-10-07/run_recovery.py--verify-only`; this check validates the preserved parent records and frozen sources without collecting new trajectories.

S12.6 Cross-collection recovery readback and two archived examples

This readback combines the 32 delivered G pairs only for describing recovery routes, retaining their original-A, original-B and supplementary-A labels. It adds no trajectory and changes no frozen score. The manuscript source package supplies `qa/recovery_costs.py`, `scripts/figure s/figure-data.json`, per-episode outputs, and `REPRODUCE-RECOVERY.md`. The script requires only the two raw ZIP archives identified in that guide, under their listed repository-relative paths; it does not require the collection-status files or X2 sources used by the separate figure-data extraction. It checks both archive hashes, event order and the before/after hashes at all 32 recovery commits. Use `--output` for a separate derived-output directory.

Example and selection. `FORMAL-G-002-B` is an original-B reuse episode selected to illustrate a recovery commit followed by a receipt query. It is not a representative latency estimate or a new experimental case. The saved input already contains the review response identifying `c1`, grant `a1`, and authorized value 3412 (proposal 3405). The handoff instead submits `c2`. Table S19 lists the subsequent bound-arm events. The matching context arm commits through the harness and makes zero additional agent tool calls; both arms complete. The bound suffix has two calls, of which only one precedes persistent recovery.

Table S19: Archived recovery example. Event suffixes belong to `FORMAL-G-002-B-bound`; identifiers and values are read from the saved events.

Events	Recorded action or feedback	Persistent effect and cost boundary
e4–e6	Harness submits `c2/a1/3412`; gate returns `INSTANCE_MISMATCH`	No new transition; recorded context and value checks pass
e10	Mismatch feedback is delivered in the next saved model input	Recovery observation begins; prior review response remains available
e11–e12	Agent commits `c1/a1/3412` successfully	Adds `transition-1`, version 3; one recovery call
e17–e18	Agent calls `get_receipt(request-2)`	Returns the same transition; before/after state hashes agree; one confirmation call

The event member is `research/agent-policy-phase2-formal-2026-10-07/formal-r esults/FORMAL-G-002-B/bound/events.jsonl` in the original `replication-package.zip`. Relative to its containing directory, `model-inputs/0.json` retains the feedback and prior grant, `state/11-before.json` and `state/11-after.json` anchor recovery, and `state/ 17-before.json` and `state/17-after.json` anchor confirmation. The source package's `tables/recovery-example-source.json` records hashes and checks for the members of both examples. The response supplies an error and the earlier review object remains available; it does not prescribe the chosen next action. A fresh authorization request is granted immediately by the deterministic reviewer in all eight fresh-authorization episodes. Neither this example nor the aggregate suffix times measure approval-queue waiting or human re-inspection.

Fresh-authorization example. `FORMAL-G-006-A` is a supplementary-A episode in which the agent requested fresh authorization although reuse of the reviewed candidate was available. The saved rejection state retains the unchanged `c1`, an active grant `a1` whose reviewed candidate is `c1` and whose authorized value is 23560, and head version 2, which the rejected request also observed;

the rejected request adds no transition. The saved input `model-inputs/0.json` already contains the proposal result creating `c1`, the `request_authorization` result returning `a1` with 23560, and the rejected `c2` request with its feedback. The first post-feedback call requests authorization for `c2`; the host issues `a2` for `c2` with the same authorized value and supersedes `a1`. Table S20 lists the subsequent bound-arm events. The context arm admits the substitute through the harness and, like the bound arm, adds two query calls; both arms complete. The bound suffix has four calls, of which two precede persistent recovery.

Table S20: Archived fresh-authorization example. Event suffixes belong to `FORMAL-G-006-A-bound`; identifiers and values are read from the saved events.

Events	Recorded action or feedback	Persistent effect and cost boundary
e4–e6	Harness submits `c2/a1/23560`; gate returns `INSTANCE_MISMATCH`	No new transition; context and value checks pass, instance check fails
e10	Mismatch feedback is delivered in the next saved model input	Recovery observation begins; the earlier review response still states `c1`, `a1` and 23560
e11–e14	Agent requests authorization for `c2`, not `c1`; host issues `a2` for `c2` and supersedes `a1`	One recovery call; the same authorized value is bound to the submitted candidate
e17–e19	Agent commits `c2/a2/23560` successfully	Adds `transition-1`, version 3; second recovery call
e23–e28	Agent calls `get_receipt(request-2)` and `get_state`	Same transition returned; two query calls after persistence

The event member is `research/agent-policy-phase2-quota-recovery-2026-10-07/results/FORMAL-G-006-A/attempt-001/bound/events.jsonl` in `quota-recovery-complete.zip`. Relative to its containing directory, `model-inputs/0.json` retains the feedback and prior grant, `state/4-after.json` holds the rejection state, and `state/17-before.json` and `state/17-after.json` anchor recovery. This episode is selected to show the mechanism, not for typicality; it does not establish why the agent preferred fresh authorization.

S13 Exploratory post hoc completed-A sensitivity analysis

This sensitivity view asks how the paired comparison changes when the supplementary A records supply the selected A task positions while the other original records remain available. It is assembled after collection and is not a replacement for the original primary estimator. Table S21 gives its provenance: the view retains all 128 original B pairs, eleven original A pairs and the final terminal supplementary attempts for the 117 selected A pairs, including their unsuccessful arms. A source map identifies each pair's collection and attempt. The original failed records remain intact outside this derived view.

Table S21: Source composition of the exploratory completed-A sensitivity view. These are selected records for existing task positions, not additional independent tasks or a newly frozen study.

Source	Pairs	Policy arms	Role in the derived view
Original A	11	22	Retain the original pairs outside the supplementary selection
Supplementary A	117	234	Use the final selected terminal attempt, retaining all failures
Original B	128	256	Retain every original B pair, including unknown coverage

The A component has 123/128 context and 125/128 bound completions; B remains at 117/128 and 119/128. Across this assembled view, context completes 240/256 and bound 244/256 arms. Averaging configuration-specific differences within each of 128 task clusters gives +1.5625 percentage points, with exploratory 95% interval $[-1.171875, 4.296875]$. A's corresponding

difference is +1.5625 points with interval $[0, 3.90625]$ and exact McNemar $p = 0.50$; B's result is unchanged. The procedure applies 10,000 scenario-stratified task-cluster bootstrap draws, seed 20261007, retaining both policies and configurations within a task.

The two original B unknown-integrity arms remain unknown. The assembled continuity-failure difference therefore has identification endpoints -12.890625 and -12.109375 percentage points, with respective bootstrap intervals $[-13.671875, -12.5]$ and $[-12.5, -11.328125]$. They are intervals for identification endpoints, not an interval around an imputed point estimate. All uncertainty statements condition on the selected sources and do not remove collection-period differences. The export explicitly marks `formal_inference_allowed=false` and labels its role exploratory post hoc sensitivity.

The adapter and source map are retained under `research/phase2-followup-sensitivity-2026-10-07/`. The adapter routes archived pair records to the unchanged frozen exporter; it does not edit the raw records or combine them into a new collection. From the repository root, run

```
python research/phase2-followup-sensitivity-2026-10-07/sensitivity.py
  --output fresh-sensitivity-output
```

The displayed lines form one command. The exported source map, tables and summary establish which original or supplementary record supplies every task position. No new model call is made.